

Hongdao Meng

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EDUCATION

New York University

Master of Science in Computer Science | GPA: 3.83

Sep. 2024 - May 2026 (Expected)

New York, NY

Beijing University of Technology

Bachelor of Engineering in Information Security | GPA: 4.00

Sep. 2020 - Jul. 2024

Beijing, CN

PROFESSIONAL & RESEARCH EXPERIENCE

TikTok, Recommendation System Infra

May 2025 – Sep. 2025

Software Engineer Intern

Seattle, WA

- Developed a Python-based **DAG-DSL** migration toolkit (leveraging **Protobuf** and **regex**) to auto-convert legacy configuration files of the **Bytedance Feature Service (BFS)**, a scalable feature extraction service powering TikTok's recommendation system, into standardized DSL definitions, boosting operator migration throughput by 85%.
- Designed and implemented a **Blade**-based **C++** operator migration pipeline for BFS, streamlining the workflow to improve operator migration speed by 83.3%, and personally migrated 20 operators.
- Architected a one-stop workflow for **User Data Accessor** covering service build, auto-generate **Python** operators, **DSL** compilation, and **RPC** request, enabling collaboration among dozens of engineers and accelerating development by 73.3%.

C2SMARTER Center

Feb. 2025 – Present

Machine Learning Engineer Intern

New York, NY

- Led a 6-member team to develop a RAG-based chatbot system using **LangChain** and **Flask-React**, achieving 24.3% accuracy improvement on the **MS MARCO** dataset ($F1=0.86$) and 33.7% faster response latency through query optimization.
- Built a hybrid retrieval framework with **MilvusDB** vector database and **BGE-M3** embeddings, improving search relevance by 21.7% and boosting query performance by 25.6% through a unified re-ranking architecture.
- Implemented a **Docker**-based data pipeline with **MongoDB** on **AWS EC2**, reducing deployment setup time by 15.7% and deploying **CI/CD** pipelines with **Jenkins** to ensure high availability.

DeepFake Detection Startup

Sep. 2024 – Dec. 2024

Machine Learning Engineer & Founder

New York, NY

- Led a 5-member team to develop core modules of a deepfake detection web platform using **React** and **TypeScript**, enabling 1,200+ concurrent users and reducing client-side rendering latency by 21.3%.
- Fine-tuned **Vision Transformer (ViT)** and **VGG16** models from **Hugging Face** for image and audio deepfake detection, achieving 91.2% and 88.1% accuracy. Built datasets with **OpenCV** and **FFmpeg**, and deployed models on **AWS EC2**.
- Built a real-time communication layer using **Django** and **WebSocket**, reducing task completion time by 25.6% with 180ms P95 latency, and deployed backend services on **Kubernetes** with **AWS ELB**, achieving 99.5% availability under 5k RPM.
- Optimized **PostgreSQL** query execution via composite index tuning, reducing average response time by 18%.

Data Mining & Security Lab

Sep. 2022 – Jul. 2024

Machine Learning Engineer Intern

Beijing, China

- Led research on **Federated Multi-View Multi-Label Learning**. Published first-author paper in *IEEE Transactions on Big Data* (2025): "**Federated Multi-View Multi-Label Classification**" (DOI: 10.1109/TBDDATA.2024.3522812).
- Proposed the **FMVML** framework enabling cross-view feature fusion and multi-label semantic classification, outperforming state-of-the-art methods with +8.3% Average Precision and -14% One Error.
- Utilized **Python/PyTorch** for model development and **Matlab** for signal processing; implemented data pipelines with **Pandas/Scikit-Learn**; prepared publication-ready documents with $\text{\LaTeX}$.

SKILLS

Languages: C/C++, Java, Python, Go, SQL, JavaScript, HTML/CSS, Shell, PHP, $\text{\LaTeX}$

Machine Learning: PyTorch, LangChain, Tensorflow, Pandas, Scikit-Learn, VGG16, ViT, Multi-View, Multi-Label

Frameworks: Blade, React, Angular, Vue.js, Django, Flask, Node.js, Spring Boot, Hadoop, Spark

Database: MySQL, Redis, MongoDB, PostgreSQL, DynamoDB, Oracle, Firebase, RocketMQ, Elasticsearch, MilvusDB

Tools: Git, Docker, AWS, Azure, CMake, Postman, CI/CD, Jenkins, Nginx, FFmpeg, OpenCV, Jira, Figma

PUBLICATION

- First Author: "Federated Multi-View Multi-Label Classification." *IEEE Transactions on Big Data*, 2025.
- Co-Author: "Susceptibility genes identification and risk evaluation model construction by transcriptome-wide association analysis for salt sensitivity of blood pressure: the EpiSS study." *BMC Genomics*, 2024.